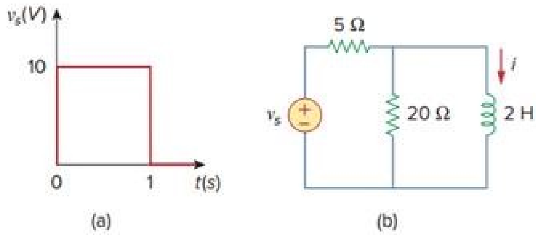
Problem

If the input pulse in Fig. 7.130(a) is applied to the circuit in Fig. 7.130(b), determine the response $i(t)$.

Fig. 7.130:

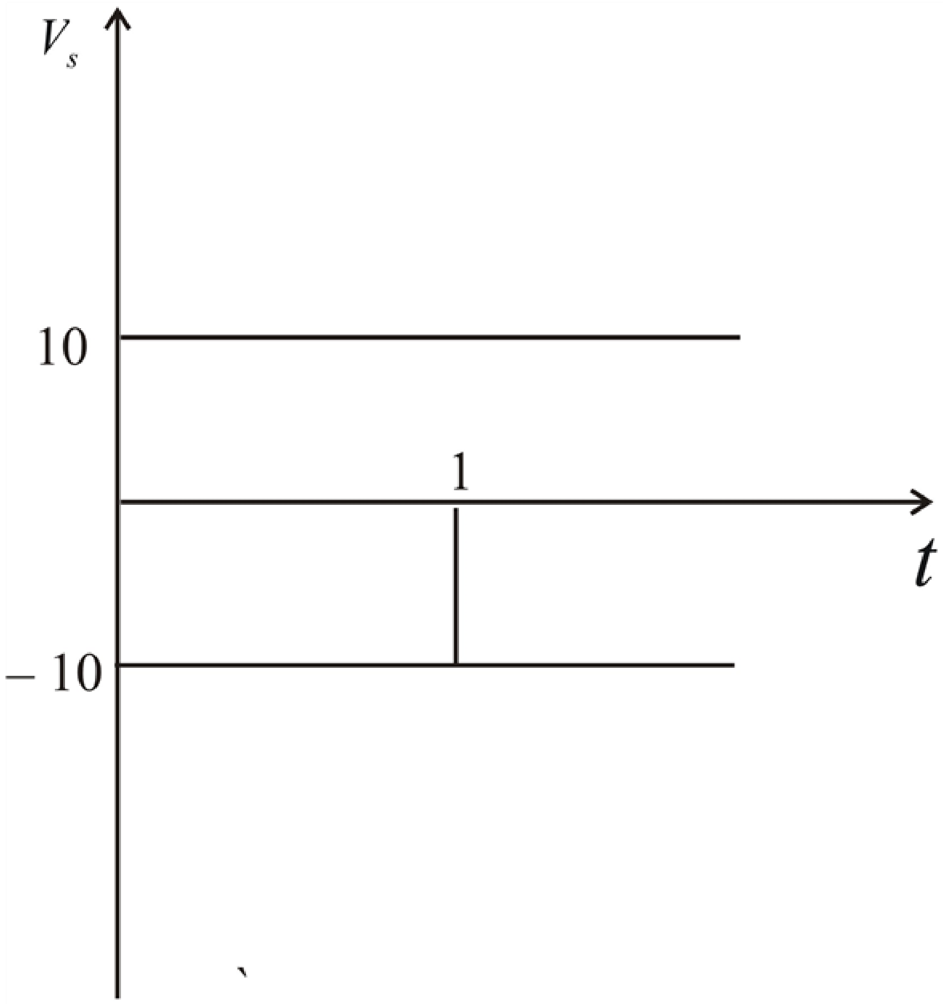


Step-by-step solution

Step 1 of 3

Since $V_S = 10[u(t) - u(t-1)$, this is same as saying that a 10 V source is through on at t = 0 and a -10 V source is turned on later at t = 1,

This is shown in figure below.



Step 2 of 3

For $0 < t < 1$, $i(0) = 0$, $i(\infty) = \frac{10}{5} = 2$

$$R_{TH} = 5 \,||\, 20 = 4, \quad \tau = \frac{L}{R_{TH}} = \frac{2}{4} = \frac{1}{2}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{\frac{-t}{\tau}}$$

$$i(t) = 2\left(1 - e^{-2t}\right)A$$

$$i(1) = 2\left(1 - e^{-2}\right) = 1.729$$

For t > 1, i(∞) = 0, Since VS = 0.

$$i(t) = i(1)\, e^{\frac{-(t-1)}{\tau}}$$

$$i(t) = 1.729\, e^{-2(t-1)}\ A$$

Step 3 of 3

Thus,

$$i(t) = \begin{cases} 2\left(1 - e^{-2t}\right)A & 0 < t < 1 \\ 1.729\, e^{-2(t-1)}\ A & t > 1 \end{cases}$$